SMART INDIA HACKATHON 2026

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TITLE PAGE

- **Problem Statement ID –** 26166
- **Problem Statement Title –** Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)
- **Theme –** Space Technology
- **PS Category –** Software
- **Team ID –** 800C4B
- **Team Name –** Bugs Janta Party

SETU

Geometry for scale and viewpoint. Physics for the Sun. Learning only for what is left.

Live demo https://arya-patil686.github.io/setu-sih2026/ **Code** https://github.com/Arya-Patil686/setu-sih2026

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SETU: spend the geometry, remove the Sun

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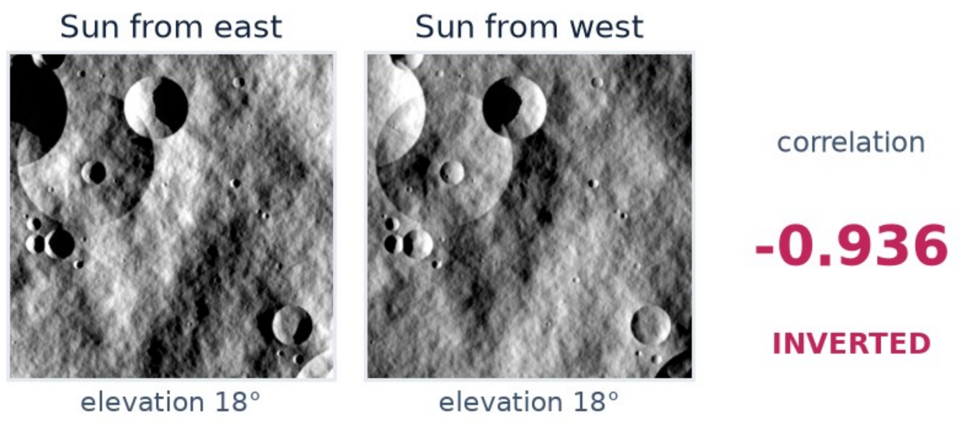
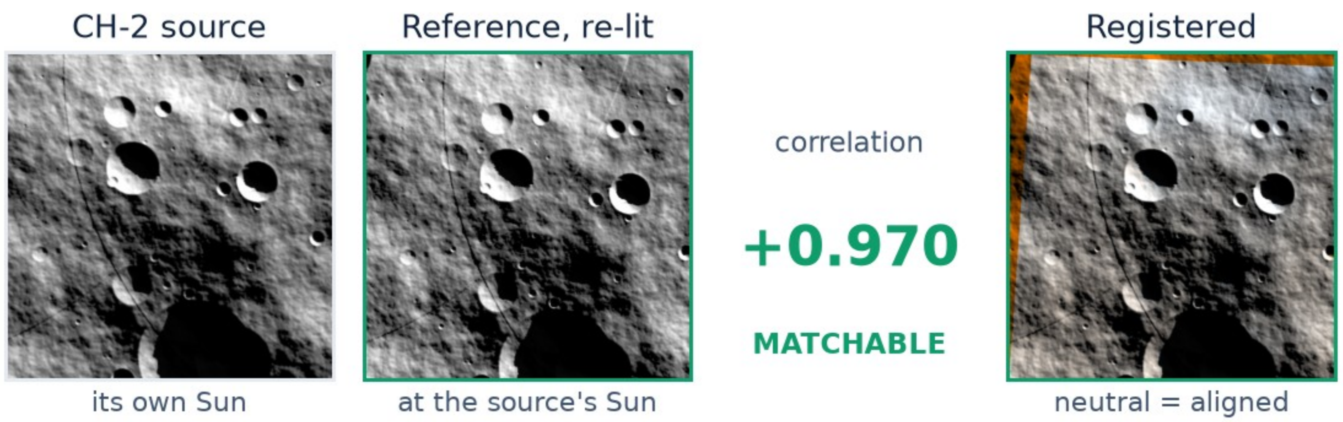
THE PROBLEM



WHAT SETU DOES



MULTI-MODAL	SUN ANGLE	SCALE	VIEWPOINT
OHRC · TMC · IIRS	any Sun, any date	up to 160x	tens of degrees
↓	↓	↓	↓
PHYSICS + STRUCTURE	**PHYSICS**	**GEOMETRY**	**GEOMETRY**
pseudo-pan over the reflected-solar window	re-render at the source's own Sun	ortho-project both at one GSD	one map projection on SLDEM2015

MEASURED ON 31 PAIRS WITH EXACT GROUND TRUTH

0.361 px	100%	416×
tie-point RMSE vs truth	matches within 3 px	better than SIFT

Multi-modal, the hardest of the three. On an IIRS-class sensor gap at a 4× scale ratio SETU reaches 0.70 px in the source frame (14 m on the ground). It is the only method that registers these pairs at all: the best baseline is 311 px.

Every match carries a 2×2 covariance, which weights the fit and ships in the tie-point file.

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TECHNICAL APPROACH

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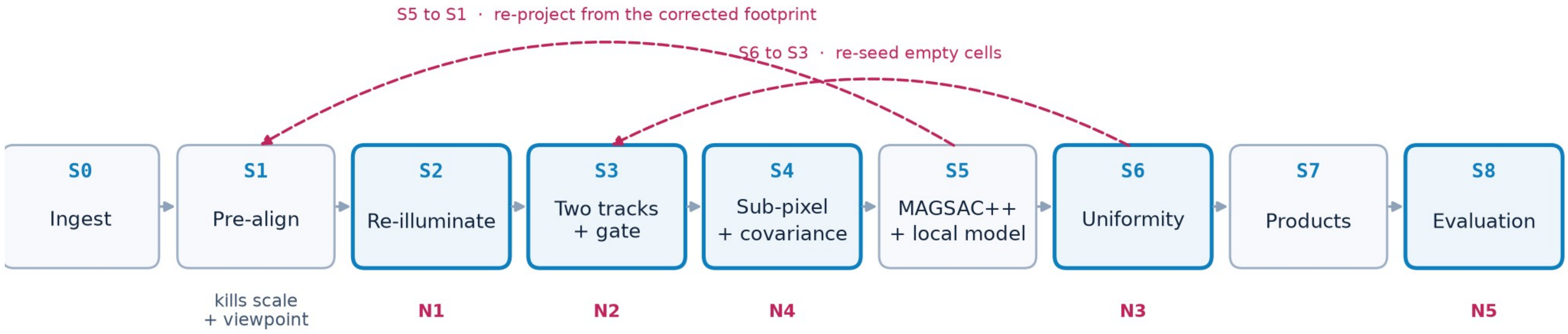

Raster / geodesy	NumPy, SciPy, rasterio, GDAL, pyproj
Planetary I/O	pds4_tools, pvl, spiceypy
Classical CV	OpenCV, MAGSAC++, scikit-image
Structural	phase congruency, MIM, CFOG, LNIFT
Deep matching	PyTorch, kornia, LoFTR, LightGlue, XFeat
Serve / ship	FastAPI, React, Vite, Docker, CI

THREE CHOICES THAT MATTER

- **Never upsample the coarser image.** Work at the coarser GSD, reach it by area-averaging.
- **Parallax is not optional off-nadir.** At 25° and 200 m of relief it is 373 OHRC pixels.
- **Harness built before the matcher.** No tuning decision was made against intuition.

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FEASIBILITY AND VIABILITY

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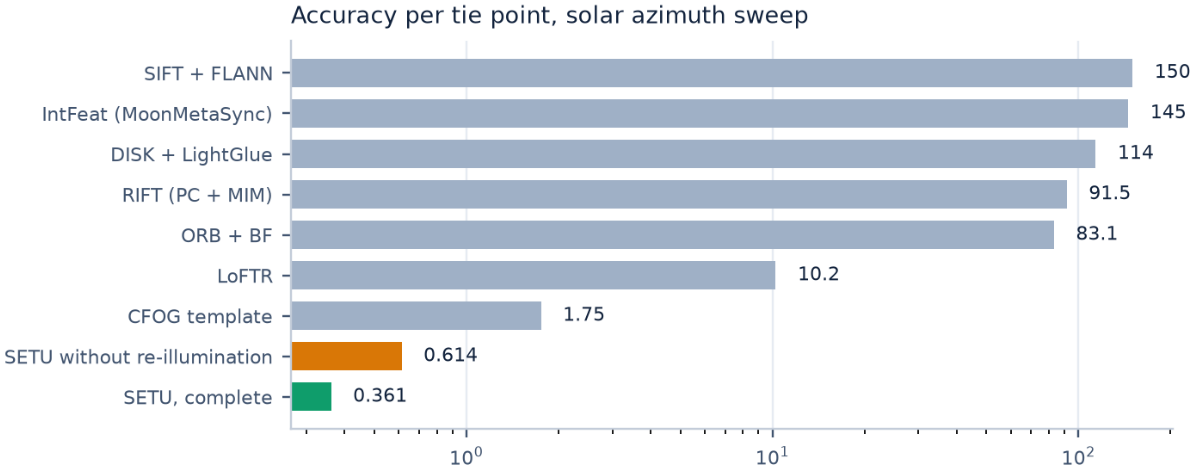
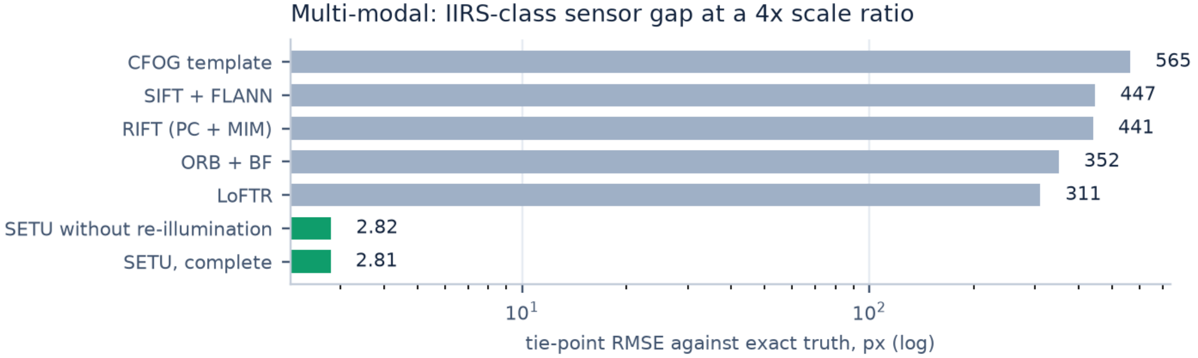
AT AN 8× SCALE RATIO, SETU DECLINES RATHER THAN LIES

144 → **0.0%** → **0** → **no registration returned**

confident matches from the deep matcher · of them correct · median error 92 px · accepted by the agreement gate · refusing to answer beats a plausible wrong answer





RISK	MITIGATION
PRADAN needs an account, archive is 200 GB	Built against the controlled benchmark; readers written and tested
SPICE / ISIS integration can eat days	Tier B corner fit is the default; Tier A degrades and says so
59 m DEM cannot render 25 cm OHRC photorealistically	Render constrains mid-frequency; sub-pixel comes from the real pair
Repetitive craters make deep matchers confidently wrong	The agreement gate, measured at 0.0% precision rejected

BUILT, NOT DESCRIBED

9 / 9 stages, both edges

41 tests, green

5.6 s per pair, no GPU

Beyond the design target it declines again. On thermal-regime bands, which the pseudo-pan window exists to avoid, every baseline returns garbage at 0% precision and SETU registers none of the 10 pairs.

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IMPACT AND BENEFITS

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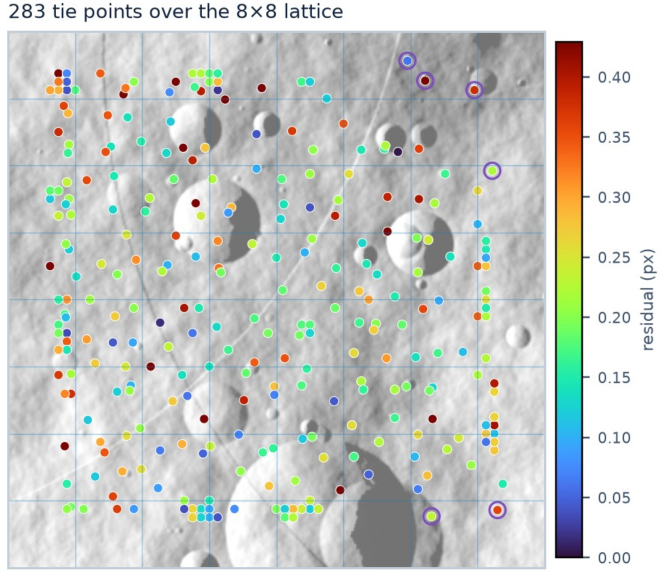
0.361 px
tie-point RMSE vs exact truth

100%
matches within 3 px of truth

100%
inlier ratio after MAGSAC++

0.81
coverage, mean over 31 pairs

416×
better per point than SIFT



coverage 0.95 on this run · purple ring = re-seeded cell

WHAT THIS BUYS ISRO

- **An archive that co-registers.** Kilometre-level geolocation is why OHRC, TMC-2 and IIRS are hard to combine.
- **A tie-point list that can be ingested.** Per-point covariance, so a bundle adjustment can weight it.
- **Pointing stability, for free.** The per-row jitter spline falls out of registration.
- **Generalises past the Moon.** Any airless body with a shape model has the same problem.

TRY IT LIVE, AND READ THE CODE

https://arya-patil686.github.io/setu-sih2026/

https://github.com/Arya-Patil686/setu-sih2026

Three registrations at increasing difficulty, the pipeline played stage by stage, tie points coloured by residual, and the full leaderboard.

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RESEARCH AND REFERENCES

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RESEARCH THIS BUILDS ON

- **He et al.** MatchAnything, TPAMI 2026, arXiv:2501.07556. Track A weights.
- **Li, Hu, Ai.** RIFT, IEEE TIP 29:3296, 2020. Track B descriptor.
- **Ye et al.** HOPC, IEEE TGRS 55(5) 2017; CFOG, ISPRS 2019.
- **Kovesi.** Image Features from Phase Congruency, 1999. Vendored.
- **Kumar et al.** MoonMetaSync, arXiv:2410.11118, 2024. The lunar floor we beat.
- **Tungathurthi.** arXiv:2602.14993, 2026. Documents the 4 to 6 km OHRC error.
- **Barker et al.** SLDEM2015, Icarus 273:346, 2016. The shape model.
- **Chowdhury et al.** OHRC in-orbit performance, Current Science 118(4):560, 2020.
- **Verma et al.** IIRS thermal emission correction, Icarus 383:115075, 2022.
- **Barath et al.** MAGSAC++, CVPR 2020. With an adaptive threshold from our covariances.

DATA

Chandrayaan-2 L1 via PRADAN · LRO NAC and NAC DTMs via ODE · SLDEM2015 at 512 ppd · Kaguya TC via JAXA DARTS · LROC WAC

LIVE DEMO

https://arya-patil686.github.io/setu-sih2026/

https://github.com/Arya-Patil686/setu-sih2026

REPRODUCE EVERY NUMBER

```
python experiments/run_sweeps.py
python scripts/build_demo_bundle.py
```

Ground truth is exact by construction: both images of every pair are rendered from one terrain model under a known transform. Results are on that benchmark; the PDS4 and PDS3 readers and the SPICE path await archive access, not code.